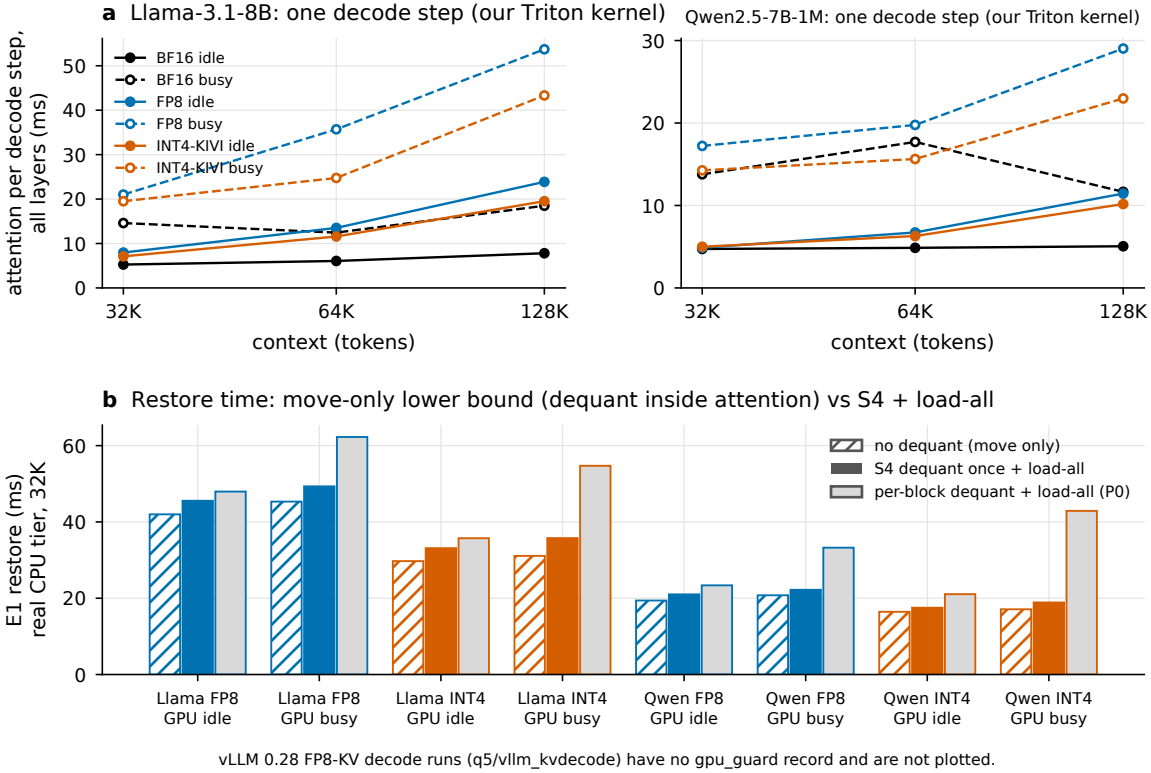
a Llama-3.1-8B: one decode step (our Triton kernel)
Qwen2.5-7B-1M: one decode step (our Triton kernel)
attention per decode step, all layers (ms)
BF16 idle
BF16 busy
FP8 idle
FP8 busy
INT4-KIVI idle
INT4-KIVI busy
0
10
20
30
40
50
32K
64K
128K
context (tokens)
b Restore time: move-only lower bound (dequant inside attention) vs S4 + load-all
E1 restore (ms) real CPU tier, 32K
no dequant (move only)
S4 dequant once + load-all
per-block dequant + load-all (P0)
0
20
40
60
Llama FP8 GPU idle
Llama FP8 GPU busy
Llama INT4 GPU idle
Llama INT4 GPU busy
Qwen FP8 GPU idle
Qwen FP8 GPU busy
Qwen INT4 GPU idle
Qwen INT4 GPU busy
vLLM 0.28 FP8-KV decode runs (q5/vllm_kvdecode) have no gpu_guard record and are not plotted.